

Lesson 2: Point Estimation; Bias, Variance, and MSE

BSTA 551: Statistical Inference

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Lesson 2: Point Estimation

Review: Where We Left Off

Key concepts from Lesson 1:

- Population vs. sample; parameters vs. statistics
- Sampling distributions
- $E(\bar{X}) = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n$
- Numerical optimization with `optimize()`

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Today's Goals:

- Define point estimators formally
- Understand bias and how to calculate it
- Learn about standard error and precision
- Introduce Mean Squared Error (MSE)
- Explore the bias-variance tradeoff

Point Estimation: Core Concepts

What is a Point Estimator? (Devore 7.1)

! Definitions

- A **parameter** is a fixed (but unknown) characteristic of a population (e.g., μ , σ , p)
- An **estimator** is a rule/formula for calculating an estimate from sample data
- An **estimate** is the actual number you calculate from a specific sample

What is a Point Estimator? (Devore 7.1)

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Example:

Concept	Symbol	Example
Parameter	μ	True mean BP reduction
Estimator	$\bar{X} = \frac{1}{n} \sum X_i$	The formula “sample mean”
Estimate	$\bar{x} = 11.2$	The number from our data

Key Distinction: Estimator vs. Estimate

Estimator: A random variable (before data is collected)

- $\bar{X}$ is a function of random variables $X_1, \dots, X_n$
- Has a sampling distribution
- Can calculate $E(\bar{X})$, $\text{Var}(\bar{X})$

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- Just one realization from the sampling distribution

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Tip

Think of it like this: the estimator is the recipe; the estimate is what you get when you cook it with specific ingredients.

The Sampling Distribution (Revisited)

Different samples give different estimates. The **sampling distribution** describes this variability.

Sample	Sample Mean ($\bar{x}$)
1	9.90
2	10.31
3	10.03
4	10.85
5	9.17
6	9.31

The Sampling Distribution (Revisited)

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Sample	Sample Mean ($\bar{x}$)
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Each sample gives a **different estimate**, but they cluster around the true value $\mu = 10$.

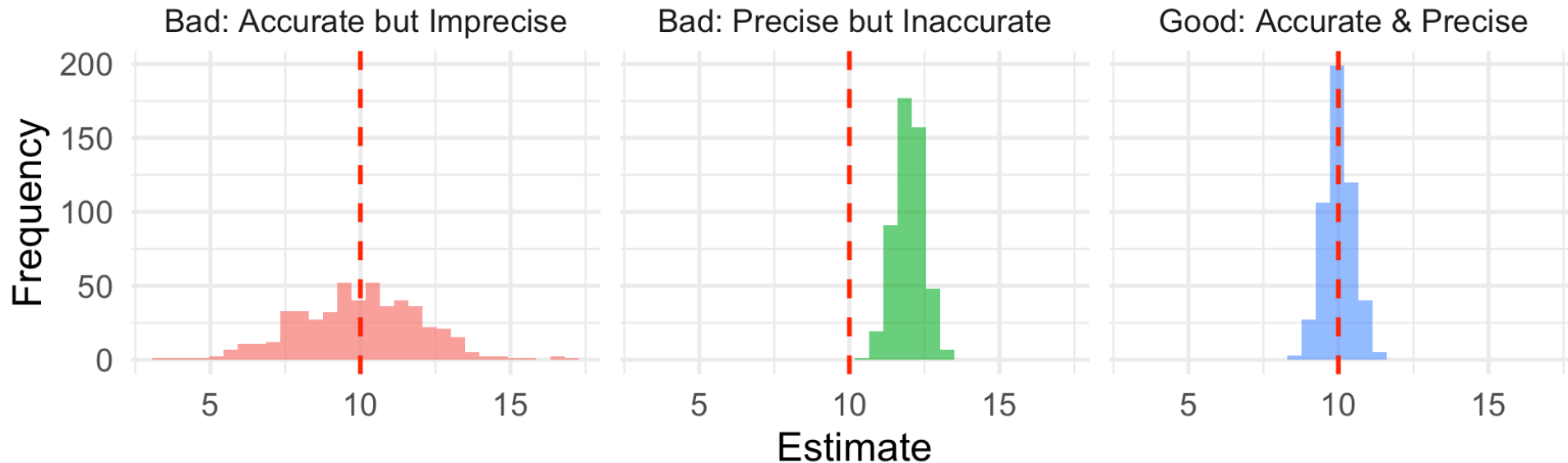
What Makes a Good Estimator?

We want estimators that are:

1. **Accurate** (unbiased): On average, hits the true value
2. **Precise** (low variance): Estimates are clustered together
3. **Efficient**: Best combination of accuracy and precision

Comparing Estimator Quality

Red line = true parameter value



Bias: Measuring Accuracy

Bias: Formal Definition

! Definition

The **bias** of an estimator $\hat{\theta}$ is:

$$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$$

An estimator is **unbiased** if $\text{Bias}(\hat{\theta}) = 0$, i.e., $E(\hat{\theta}) = \theta$.

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An estimator is **unbiased** if $\text{Bias}(\hat{\theta}) = 0$, i.e., $E(\hat{\theta}) = \theta$.

Interpretation:

- Bias measures systematic error
- Positive bias = tends to overestimate
- Negative bias = tends to underestimate
- Zero bias = correct on average

Worked Example: Proving Sample Mean is Unbiased

Claim: The sample mean $\bar{X}$ is an unbiased estimator of μ .

Proof: We need to show $E(\bar{X}) = \mu$.

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We already showed that $E(\bar{X}) = \mu$, so:

$$\text{Bias}(\bar{X}) = \mu - \mu = 0 \checkmark$$

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Conclusion: The sample mean is unbiased for estimating the population mean.

Worked Example: Sample Proportion

Setup: In a vaccine trial, X patients out of n develop immunity. The estimator is $\hat{p} = X/n$.

Claim: $\hat{p}$ is unbiased for the true immunity rate p .

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Proof: Since $X \sim \text{Binomial}(n, p)$, we know $E(X) = np$.

$$E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{1}{n}E(X) = \frac{1}{n} \cdot np = p$$

Worked Example: Sample Proportion

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$$\text{Bias}(\hat{p}) = E(\hat{p}) - p = p - p = 0 \checkmark$$

Concrete Calculation: Bias of Sample Proportion

Data: In a study of 80 patients, 52 showed improvement.

```
1 n <- 80
2 x <- 52
3 p_hat <- x / n
4
5 cat("Sample proportion:", p_hat, "\n")
```

Sample proportion: 0.65

```
1 cat("If true p = 0.65, what is the bias of this single estimate?\n")
```

If true $p = 0.65$, what is the bias of this single estimate?

```
1 cat("Observed - True =", p_hat - 0.65)
```

Observed - True = 0

Concrete Calculation: Bias of Sample Proportion

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If true $p = 0.65$, what is the bias of this single estimate?

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1 cat("Observed - True =", p_hat - 0.65)
```

Observed - True = 0

Important distinction:

- A single estimate can be above or below the true value
- **Bias** refers to the average behavior across many samples
- An unbiased estimator still gives wrong answers for individual samples!

Simulation: Verifying Unbiasedness

```
1 # Verify that sample proportion is unbiased
2 true_p <- 0.65
3 n_patients <- 80
4 n_simulations <- 10000
5
6 proportion_simulation <- tibble(sim = 1:n_simulations) |>
7   mutate(
8     successes = rbinom(n_simulations, size = n_patients, prob = true_p),
9     p_hat = successes / n_patients
10  )
11
12 proportion_simulation |>
13   summarize(
14     true_p = true_p,
15     mean_of_estimates = mean(p_hat),
16     empirical_bias = mean(p_hat) - true_p
17  )
```

A tibble: 1 × 3

	true_p	mean_of_estimates	empirical_bias
	<dbl>	<dbl>	<dbl>
1	0.65	0.650	-0.000182

The bias is essentially zero (just simulation noise)!



A Biased Estimator: The Maximum

Problem: Estimate the upper bound θ of a $\text{Uniform}[0, \theta]$ distribution.

Natural idea: Use the largest observation: $\hat{\theta} = \max(X_1, \dots, X_n)$

A Biased Estimator: The Maximum

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Think about it: Can this estimator ever *overestimate* θ ?

A Biased Estimator: The Maximum

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Natural idea: Use the largest observation: $\hat{\theta} = \max(X_1, \dots, X_n)$

Think about it: Can this estimator ever *overestimate* θ ?

No! The sample maximum is always $\leq$ the population maximum.

This means the estimator is **biased low**.

Calculating the Bias Mathematically

For $X_1, \dots, X_n \sim \text{Uniform}[0, \theta]$, it can be shown that:

$$E(\max(X_1, \dots, X_n)) = \frac{n}{n+1}\theta$$

Calculating the Bias Mathematically

For $X_1, \dots, X_n \sim \text{Uniform}[0, \theta]$, it can be shown that:

$$E(\max(X_1, \dots, X_n)) = \frac{n}{n+1}\theta$$

Bias calculation:

$$\text{Bias} = E(\hat{\theta}) - \theta = \frac{n}{n+1}\theta - \theta = -\frac{\theta}{n+1}$$

Calculating the Bias Mathematically

For $X_1, \dots, X_n \sim \text{Uniform}[0, \theta]$, it can be shown that:

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Bias calculation:

$$\text{Bias} = E(\hat{\theta}) - \theta = \frac{n}{n+1}\theta - \theta = -\frac{\theta}{n+1}$$

Example: If $\theta = 10$ and $n = 5$:

$$\text{Bias} = -\frac{10}{6} = -1.67$$

The estimator underestimates by about 1.67 on average.

Your Turn: Calculate Bias

Exercise: A lab instrument has a maximum detection limit θ . We take $n = 9$ measurements from $\text{Uniform}[0, \theta]$ and use the maximum as our estimate.

1. If $\theta = 100$, what is $E(\hat{\theta})$?
2. What is the bias?
3. By what percentage does this estimator underestimate on average?

Your Turn: Calculate Bias

Exercise: A lab instrument has a maximum detection limit θ . We take $n = 9$ measurements from $\text{Uniform}[0, \theta]$ and use the maximum as our estimate.

1. If $\theta = 100$, what is $E(\hat{\theta})$?
2. What is the bias?
3. By what percentage does this estimator underestimate on average?

Solution:

1. $E(\hat{\theta}) = \frac{9}{10} \times 100 = 90$
2. $\text{Bias} = 90 - 100 = -10$
3. Underestimates by $\frac{10}{100} = 10\%$

Simulation: Visualizing the Biased Estimator

```
1 true_theta <- 100
2 n <- 9
3 n_sims <- 5000
4
5 max_simulation <- tibble(sim = 1:n_sims) |>
6   mutate(
7     max_estimate = map_dbl(sim, \(s) max(runif(n, 0, true_theta)))
8   )
9
10 # Calculate empirical bias
11 max_simulation |>
12   summarize(
13     theoretical_E = n / (n + 1) * true_theta,
14     empirical_mean = mean(max_estimate),
15     theoretical_bias = -true_theta / (n + 1),
16     empirical_bias = mean(max_estimate) - true_theta
17   )
```

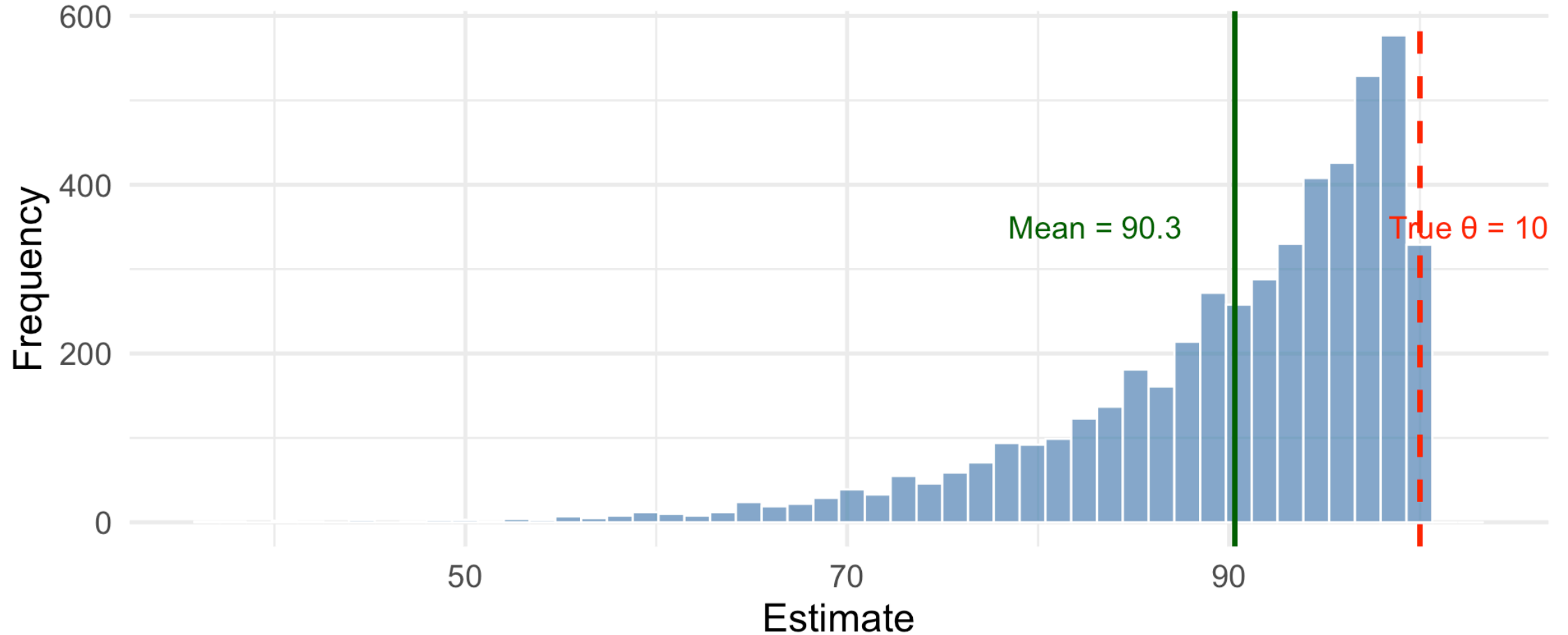
Simulation: Visualizing the Biased Estimator

```
# A tibble: 1 × 4
  theoretical_E empirical_mean theoretical_bias empirical_bias
    <dbl>         <dbl>         <dbl>         <dbl>
1      90         90.3         -10         -9.70
```

Visualizing the Biased Estimator

Distribution of Maximum Estimator (Biased)

True $\theta = 100$, $n = 9$



Correcting the Bias

Idea: Multiply by a correction factor to “un-bias” the estimator.

Since $E(\max) = \frac{n}{n+1}\theta$, we can define:

$$\hat{\theta}_{\text{unbiased}} = \frac{n+1}{n} \cdot \max(X_1, \dots, X_n)$$

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$$\hat{\theta}_{\text{unbiased}} = \frac{n+1}{n} \cdot \max(X_1, \dots, X_n)$$

Check:

$$E(\hat{\theta}_{\text{unbiased}}) = \frac{n+1}{n} \cdot E(\max) = \frac{n+1}{n} \cdot \frac{n}{n+1}\theta = \theta \checkmark$$

Your Turn: Apply the Correction

Exercise: Using the lab instrument example with $n = 9$ and $\theta = 100$:

1. If you observe $\text{max} = 92$, what is the biased estimate?
2. What is the unbiased estimate?

Your Turn: Apply the Correction

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2. What is the unbiased estimate?

Solution:

1. Biased estimate: $\hat{\theta}_b = 92$
2. Unbiased estimate: $\hat{\theta}_u = \frac{10}{9} \times 92 = 102.2$

```
1 observed_max <- 92
2 n <- 9
3
4 biased_est <- observed_max
5 unbiased_est <- (n + 1) / n * observed_max
6
7 cat("Biased estimate:", biased_est, "\n")
```

Biased estimate: 92

```
1 cat("Unbiased estimate:", round(unbiased_est, 1))
```

Unbiased estimate: 102.2

Standard Error: Measuring Precision

Standard Error: Definition

! Definition

The **standard error** of an estimator is its standard deviation:

$$SE(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}$$

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Key Standard Errors:

Estimator	Standard Error
Sample mean $\bar{X}$	$\frac{\sigma}{\sqrt{n}}$
Sample proportion $\hat{p}$	$\sqrt{\frac{p(1-p)}{n}}$

Worked Example: Standard Error of Sample Mean

Problem: In a blood pressure study, the population SD is $\sigma = 15$ mmHg. Calculate the standard error of $\bar{X}$ for sample sizes $n = 25$ and $n = 100$.

Worked Example: Standard Error of Sample Mean

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Solution:

For $n = 25$:

$$SE(\bar{X}) = \frac{15}{\sqrt{25}} = \frac{15}{5} = 3 \text{ mmHg}$$

For $n = 100$:

$$SE(\bar{X}) = \frac{15}{\sqrt{100}} = \frac{15}{10} = 1.5 \text{ mmHg}$$

Worked Example: Standard Error of Sample Mean

Problem: In a blood pressure study, the population SD is $\sigma = 15$ mmHg. Calculate the standard error of $\bar{X}$ for sample sizes $n = 25$ and $n = 100$.

Solution:

For $n = 25$:

$$SE(\bar{X}) = \frac{15}{\sqrt{25}} = \frac{15}{5} = 3 \text{ mmHg}$$

For $n = 100$:

$$SE(\bar{X}) = \frac{15}{\sqrt{100}} = \frac{15}{10} = 1.5 \text{ mmHg}$$

Interpretation: With 100 patients, our estimate is twice as precise as with 25 patients.

Your Turn: Calculate Standard Error

Exercise: A survey measures patient satisfaction on a 0-100 scale. The population standard deviation is $\sigma = 20$.

1. What is the SE of $\bar{X}$ for $n = 16$ patients?
2. What sample size is needed to achieve $SE = 2$?

Your Turn: Calculate Standard Error

Exercise: A survey measures patient satisfaction on a 0-100 scale. The population standard deviation is $\sigma = 20$.

1. What is the SE of $\bar{X}$ for $n = 16$ patients?
2. What sample size is needed to achieve $SE = 2$?

Solutions:

1. $SE = \frac{20}{\sqrt{16}} = \frac{20}{4} = 5$

2. We need $\frac{20}{\sqrt{n}} = 2$, so $\sqrt{n} = 10$, thus $n = 100$

```
1 sigma <- 20
2 # Problem 1
3 se_n16 <- sigma / sqrt(16)
4 # Problem 2
5 n_needed <- (sigma / 2)^2
6
7 cat("SE for n=16:", se_n16, "\nSample size for SE=2:", n_needed)
```

SE for n=16: 5

Sample size for SE=2: 100

The Problem with Unknown Parameters

Issue: Standard errors often involve unknown parameters!

- SE of $\bar{X}$ requires knowing σ
- SE of $\hat{p}$ requires knowing p

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Solution: Estimated standard error — substitute estimates for unknown parameters

$$\widehat{SE}(\bar{X}) = \frac{s}{\sqrt{n}}$$

$$\widehat{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Example: Estimated Standard Error

```
1 # Blood pressure data from 25 patients
2 bp_reductions <- c(12, 8, 15, 10, 7, 14, 11, 9, 13, 16,
3                   8, 12, 10, 14, 11, 9, 15, 13, 7, 12,
4                   10, 8, 14, 11, 13)
5
6 n <- length(bp_reductions)
7 x_bar <- mean(bp_reductions)
8 s <- sd(bp_reductions)
9
10 # Estimated standard error
11 se_estimated <- s / sqrt(n)
12
13 tibble(
14   Statistic = c("Sample Mean", "Sample SD", "Sample Size", "Estimated SE"),
15   Value = c(x_bar, round(s, 2), n, round(se_estimated, 2))
16 )
```

```
# A tibble: 4 × 2
  Statistic      Value
  <chr>         <dbl>
1 Sample Mean   11.3
2 Sample SD     2.64
3 Sample Size   25
4 Estimated SE   0.52
```

Mean Squared Error

Mean Squared Error: Combining Bias and Variance

What if we have to choose between a biased estimator with low variance and an unbiased estimator with high variance?

Mean Squared Error: Combining Bias and Variance

What if we have to choose between a biased estimator with low variance and an unbiased estimator with high variance?

! Definition: Mean Squared Error

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

Key Formula:

$$\text{MSE} = \text{Variance} + \text{Bias}^2$$

Mean Squared Error: Combining Bias and Variance

What if we have to choose between a biased estimator with low variance and an unbiased estimator with high variance?

! Definition: Mean Squared Error

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

Key Formula:

$$\text{MSE} = \text{Variance} + \text{Bias}^2$$

MSE captures total error — both systematic (bias) and random (variance).

Proving the MSE Formula

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

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Add and subtract $E(\hat{\theta})$:

$$= E[(\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta)^2]$$

Proving the MSE Formula

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Add and subtract $E(\hat{\theta})$:

$$= E[(\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta)^2]$$

Expand the square:

$$= E[(\hat{\theta} - E(\hat{\theta}))^2] + 2E[(\hat{\theta} - E(\hat{\theta}))](E(\hat{\theta}) - \theta) + (E(\hat{\theta}) - \theta)^2$$

Proving the MSE Formula

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

Add and subtract $E(\hat{\theta})$:

$$= E[(\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta)^2]$$

Expand the square:

$$= E[(\hat{\theta} - E(\hat{\theta}))^2] + 2E[(\hat{\theta} - E(\hat{\theta}))](E(\hat{\theta}) - \theta) + (E(\hat{\theta}) - \theta)^2$$

The middle term equals zero because $E[\hat{\theta} - E(\hat{\theta})] = 0$:

$$= \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

Worked Example: Computing MSE

Setup: Estimating θ from $\text{Uniform}[0, \theta]$ with $n = 9$ and $\theta = 100$.

Biased estimator: $\hat{\theta}_b = \max(X_i)$

From theory:

- $E(\hat{\theta}_b) = \frac{9}{10}(100) = 90$
- $\text{Var}(\hat{\theta}_b) = \frac{n\theta^2}{(n+1)^2(n+2)} = \frac{9 \times 100^2}{100 \times 11} = 81.82$

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Biased estimator: $\hat{\theta}_b = \max(X_i)$

From theory:

- $E(\hat{\theta}_b) = \frac{9}{10}(100) = 90$
- $\text{Var}(\hat{\theta}_b) = \frac{n\theta^2}{(n+1)^2(n+2)} = \frac{9 \times 100^2}{100 \times 11} = 81.82$

$$\text{Bias} = 90 - 100 = -10$$

$$\text{MSE} = 81.82 + (-10)^2 = 81.82 + 100 = 181.82$$

Your Turn: Calculate MSE

Exercise: For the **unbiased** estimator $\hat{\theta}_u = \frac{n+1}{n} \max(X_i)$ with $n = 9$ and $\theta = 100$:

1. What is the bias?
2. If $\text{Var}(\hat{\theta}_u) = 101.01$, what is the MSE?
3. Which estimator has lower MSE: biased or unbiased?

Your Turn: Calculate MSE

Exercise: For the **unbiased** estimator $\hat{\theta}_u = \frac{n+1}{n} \max(X_i)$ with $n = 9$ and $\theta = 100$:

1. What is the bias?
2. If $\text{Var}(\hat{\theta}_u) = 101.01$, what is the MSE?
3. Which estimator has lower MSE: biased or unbiased?

Solutions:

1. Bias = 0 (it's unbiased!)
2. $\text{MSE} = 101.01 + 0^2 = 101.01$
3. Unbiased has lower MSE ($101.01 < 181.82$)

But this isn't always the case!

The Bias-Variance Tradeoff

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Sometimes a **biased** estimator has **lower MSE** than an unbiased one!

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Sometimes a **biased** estimator has **lower MSE** than an unbiased one!

Example: Estimating a proportion p with $n = 20$ observations

Estimator 1: Standard: $\hat{p}_1 = \frac{X}{n}$ (unbiased)

Estimator 2: “Add-two”: $\hat{p}_2 = \frac{X+2}{n+4}$ (biased toward 0.5)

Comparing Proportion Estimators: Theory

For $\hat{p}_1 = X/n$ (standard):

- Bias = 0
- Variance = $\frac{p(1-p)}{n}$
- MSE = $\frac{p(1-p)}{n}$

Comparing Proportion Estimators: Theory

For $\hat{p}_1 = X/n$ (standard):

- Bias = 0
- Variance = $\frac{p(1-p)}{n}$
- MSE = $\frac{p(1-p)}{n}$

For $\hat{p}_2 = \frac{X+2}{n+4}$ (add-two):

- Bias = $\frac{2-4p}{n+4}$
- Variance = $\frac{np(1-p)}{(n+4)^2}$
- MSE = Variance + Bias²

Your Turn: Calculate Bias of Add-Two Estimator

Exercise: For $n = 20$ and $p = 0.3$:

1. Calculate the bias of $\hat{p}_2 = \frac{X+2}{n+4}$

Hint: $E(X) = np$ for binomial, so $E(\hat{p}_2) = \frac{np+2}{n+4}$

Your Turn: Calculate Bias of Add-Two Estimator

Exercise: For $n = 20$ and $p = 0.3$:

1. Calculate the bias of $\hat{p}_2 = \frac{X+2}{n+4}$

Hint: $E(X) = np$ for binomial, so $E(\hat{p}_2) = \frac{np+2}{n+4}$

Solution:

$$E(\hat{p}_2) = \frac{20(0.3) + 2}{24} = \frac{8}{24} = 0.333$$

$$\text{Bias} = 0.333 - 0.3 = 0.033$$

The add-two estimator is biased **toward 0.5** (and 0.333 is closer to 0.5 than 0.3 is).

Simulation: Comparing the Estimators

```
1 true_p <- 0.3
2 n <- 20
3 n_sims <- 10000
4
5 # Simulate both estimators
6 comparison_sim <- tibble(sim = 1:n_sims) |>
7   mutate(
8     x = rbinom(n_sims, size = n, prob = true_p),
9     p_hat_standard = x / n,
10    p_hat_addtwo = (x + 2) / (n + 4)
11  )
12
13 # Compare MSE
14 comparison_sim |>
15   summarize(
16     `Standard Bias` = mean(p_hat_standard) - true_p,
17     `Add-Two Bias` = mean(p_hat_addtwo) - true_p,
18     `Standard Variance` = var(p_hat_standard),
19     `Add-Two Variance` = var(p_hat_addtwo),
20     `Standard MSE` = mean((p_hat_standard - true_p)^2),
21     `Add-Two MSE` = mean((p_hat_addtwo - true_p)^2)
22  ) |>
23   pivot_longer(everything(), names_to = "Metric", values_to = "Value") |>
```


Simulation: Comparing the Estimators

```
1 true_p <- 0.3
2 n <- 20
3 n_sims <- 10000
4
5 # Simulate both estimators
6 comparison_sim <- tibble(sim = 1:n_sims) |>
7   mutate(
8     x = rbinom(n_sims, size = n, prob = true_p),
9     p_hat_standard = x / n,
10    p_hat_addtwo = (x + 2) / (n + 4)
11  )
12
13 # Compare MSE
14 comparison_sim |>
15   summarize(
16     `Standard Bias` = mean(p_hat_standard) - true_p,
17     `Add-Two Bias` = mean(p_hat_addtwo) - true_p,
18     `Standard Variance` = var(p_hat_standard),
19     `Add-Two Variance` = var(p_hat_addtwo),
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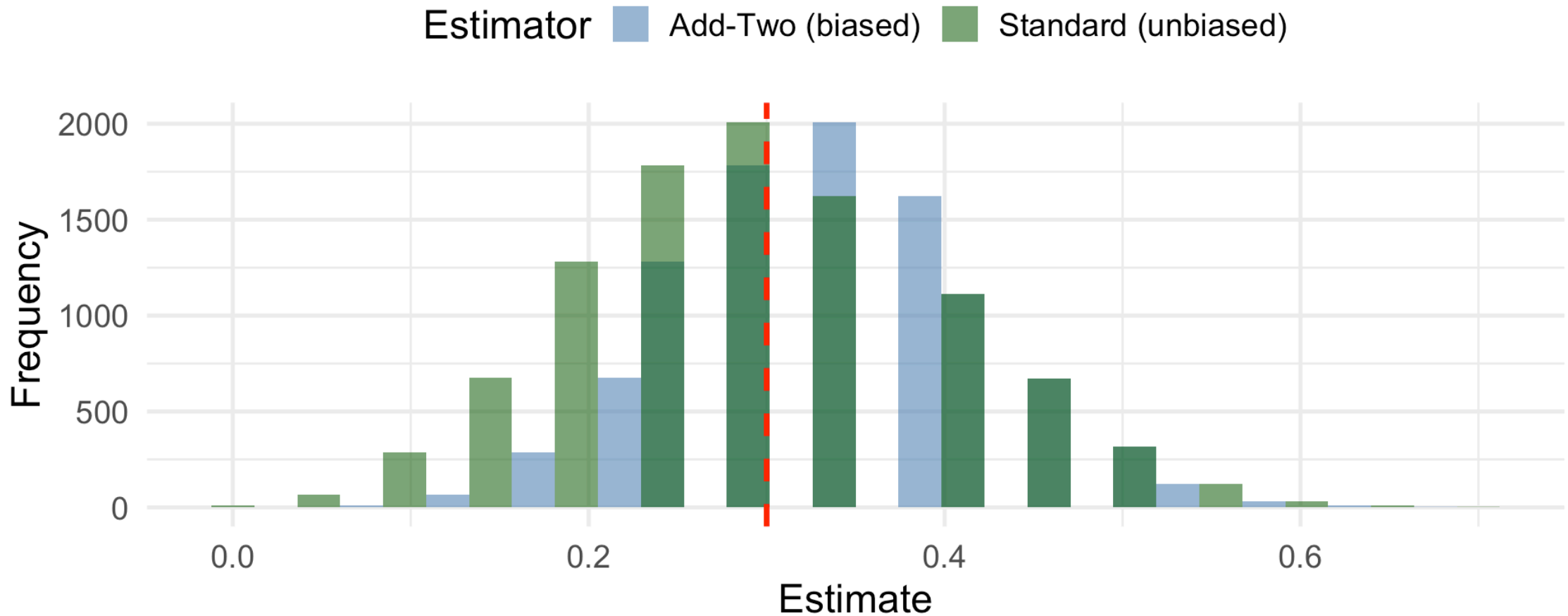
Simulation: Comparing the Estimators

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```

Visualizing the Tradeoff

Comparing Two Proportion Estimators

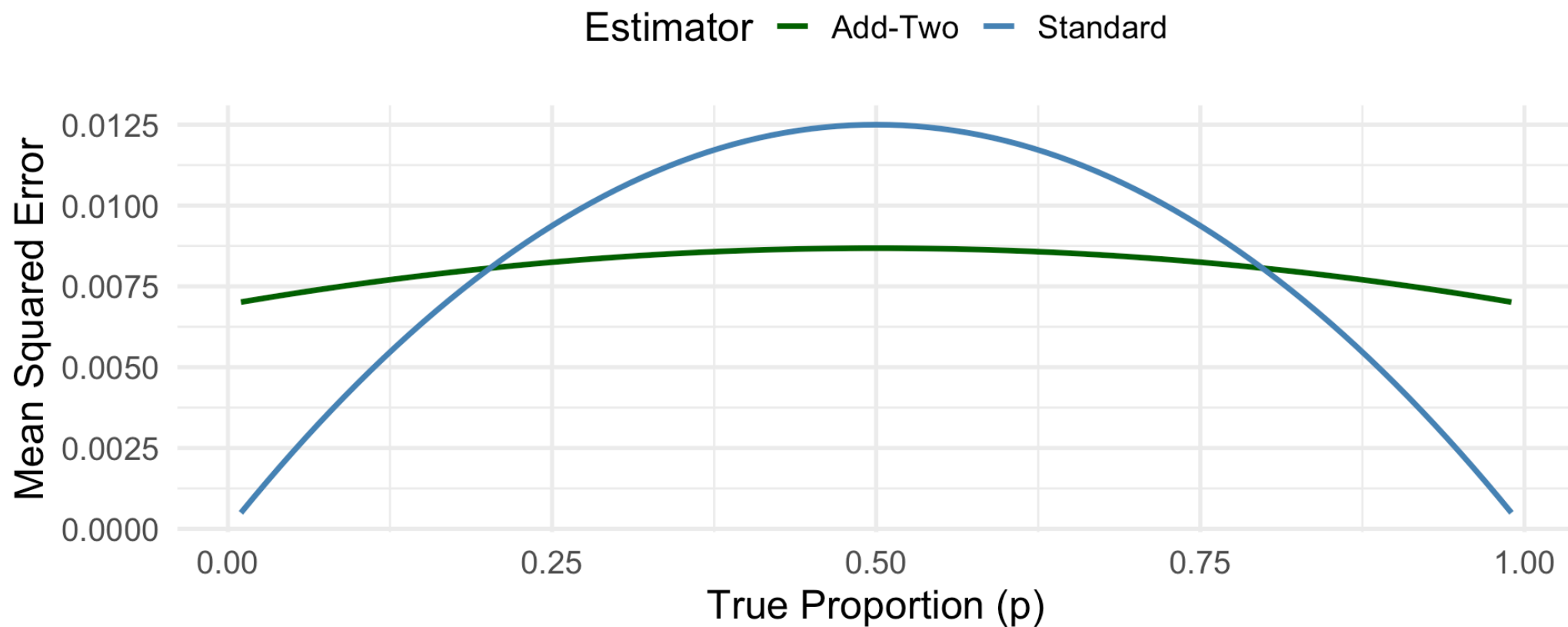
True $p = 0.3$, $n = 20$



MSE Comparison Across Different True Values

MSE Comparison: Which Estimator is Better?

Neither dominates everywhere — the winner depends on true p



Key Insight: The “best” estimator depends on the true parameter value!

Medical Application: Disease Prevalence

Scenario: Estimating prevalence of a rare disease ($p \approx 0.05$) vs. a common condition ($p \approx 0.5$).

```
1 # Compare MSE at different prevalence levels
2 n <- 50
3
4 mse_at_p <- function(p, n) {
5   mse_std <- p * (1-p) / n
6   bias_add2 <- (2 - 4*p) / (n + 4)
7   var_add2 <- n * p * (1-p) / (n + 4)^2
8   mse_add2 <- var_add2 + bias_add2^2
9
10  tibble(p = p, MSE_Standard = mse_std, MSE_AddTwo = mse_add2,
11         Better = ifelse(mse_std < mse_add2, "Standard", "Add-Two"))
12 }
13
14 bind_rows(
15   mse_at_p(0.05, n),
16   mse_at_p(0.50, n)
17 ) |>
18 mutate(across(where(is.numeric), \(x) round(x, 5)))
```

A tibble: 2 × 4

p	MSE_Standard	MSE_AddTwo	Better
---	--------------	------------	--------

<dbl>	<dbl>	<dbl>	<chr>
-------	-------	-------	-------



Sample Variance: Why n-1?

Two formulas for sample variance:

$$S^2 = \frac{\sum (X_i - \bar{X})^2}{n - 1} \quad \text{vs.} \quad \tilde{S}^2 = \frac{\sum (X_i - \bar{X})^2}{n}$$

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Question: Why do we divide by $n - 1$ instead of n ?

Answer: Dividing by n gives a biased estimator!

Simulation: Comparing Variance Estimators

```
1 true_variance <- 100 #  $\sigma^2 = 100$ 
2 n <- 10
3 n_sims <- 10000
4
5 variance_sim <- tibble(sim = 1:n_sims) |>
6   mutate(
7     sample_data = map(sim, \(s) rnorm(n, 0, sqrt(true_variance))),
8     s2_n_minus_1 = map_dbl(sample_data, var),
9     s2_n = map_dbl(sample_data, \(x) sum((x - mean(x))^2) / n)
10  )
11
12 variance_sim |>
13   summarize(
14     `True  $\sigma^2` = true_variance,
15     `E[S^2 with n-1]` = mean(s2_n_minus_1),
16     `E[S^2 with n]` = mean(s2_n),
17     `Bias (n-1)` = mean(s2_n_minus_1) - true_variance,
18     `Bias (n)` = mean(s2_n) - true_variance
19   ) |>
20   mutate(across(where(is.numeric), \(x) round(x, 2)))$ 
```

A tibble: 1 × 5

`True  $\sigma^2` `E[S^2 with n-1]` `E[S^2 with n]` `Bias (n-1)` `Bias (n)`$



Your Turn: Comprehensive Example

Exercise: A clinical trial measures cholesterol reduction. Based on $n = 36$ patients:

- Sample mean: $\bar{x} = 25$ mg/dL
- Sample SD: $s = 12$ mg/dL

Calculate:

1. The estimated standard error of $\bar{X}$
2. If the true mean reduction is $\mu = 24$, and we repeated this trial many times, what would be the expected MSE of $\bar{X}$?

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- Sample SD: $s = 12$ mg/dL

Calculate:

1. The estimated standard error of $\bar{X}$
2. If the true mean reduction is $\mu = 24$, and we repeated this trial many times, what would be the expected MSE of $\bar{X}$?

Solutions:

1. $\widehat{SE} = \frac{12}{\sqrt{36}} = 2$ mg/dL

2. $\bar{X}$ is unbiased, so $MSE = \text{Var}(\bar{X}) = \frac{\sigma^2}{n} \approx \frac{144}{36} = 4$

Summary and Looking Ahead

Putting It All Together: Estimator Summary

Property	Formula	Interpretation
Bias	$E(\hat{\theta}) - \theta$	Systematic error
Variance	$E[(\hat{\theta} - E(\hat{\theta}))^2]$	Random variability
Std Error	$\sqrt{\text{Var}(\hat{\theta})}$	Typical deviation
MSE	$\text{Var} + \text{Bias}^2$	Total error

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Guidelines for Choosing Estimators

1. Unbiasedness is desirable but not always essential
2. Lower variance/SE means more precision
3. MSE provides a single criterion combining both
4. Sometimes biased estimators have lower MSE!

Lesson 2 Summary

Key Concepts:

1. **Point Estimators:** Rules for calculating estimates from data

- Estimator = random variable; estimate = specific value

2. **Bias:** $\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$

- Unbiased if $E(\hat{\theta}) = \theta$
- Can sometimes correct biased estimators

3. **Standard Error:** $SE(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}$

- Measures precision of the estimator

4. **Mean Squared Error:** $\text{MSE} = \text{Var} + \text{Bias}^2$

- Captures total estimation error
- Bias-variance tradeoff: sometimes biased is better!

Lesson 2 Practice Problems

1. For a Uniform $[0, \theta]$ distribution with $n = 20$ observations and $\theta = 50$, calculate:
 - The expected value of the maximum
 - The bias of using the maximum as an estimator
 - The corrected unbiased estimator
2. Using simulation, compare the MSE of the standard proportion estimator vs. the add-two estimator for $n = 10$ and $p = 0.1, 0.3, 0.5$.
3. For a sample of size n from Exponential(λ), the MLE is $\hat{\lambda} = 1/\bar{X}$. It can be shown that $E(\hat{\lambda}) = \frac{n}{n-1} \lambda$. Calculate the bias and propose an unbiased estimator.

Next Week Preview

Week 2: Minimum Variance Unbiased Estimators

- Among all unbiased estimators, which has smallest variance?
- The Cramér-Rao lower bound
- Efficiency of estimators
- Introduction to Maximum Likelihood Estimation

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Chapter 7.1
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Chapter 6.

Questions?

Thank you!